

ST ATHAN, United Kingdom

WMO: 037160

Lat: 51.4053N

Lon: 3.4408W

Elev: 50

StdP: 100.73

Time Zone: 0.00 (EUW)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.2	-0.8	-5.9	2.3	-0.3	-4.5	2.6	0.7	25.1	9.0	22.7	9.3	4.1	70	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.2	24.1	17.9	22.1	17.0	20.6	16.3	18.8	22.3	17.9	20.5	17.2	19.5	4.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.6	19.8	16.9	12.1	19.0	16.2	11.6	18.4	53.3	22.4	50.6	20.6	48.5	19.6	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.3	12.3	10.5	-4.3	27.5	1.7	2.2	-5.5	29.1	-6.5	30.3	-7.4	31.6	-8.7	33.1		
(5)			-5.1	20.9	1.6	1.8	-6.2	22.2	-7.2	23.3	-8.1	24.3	-9.3	25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	5.8	5.6	7.1	9.5	12.1	15.1	16.7	16.4	14.8	11.9	8.6	6.4	(6)
(7)		DBStd	4.80	3.12	2.84	2.61	2.50	2.41	2.13	2.21	1.75	2.07	2.68	3.00	3.35	(7)
(8)		HDD10.0	581	131	125	93	38	6	0	0	0	13	59	116		(8)
(9)		HDD18.3	2746	388	357	348	266	193	101	61	63	108	199	292	370	(9)
(10)		CDD10.0	897	1	1	3	22	71	153	209	199	143	73	17	4	(10)
(11)		CDD18.3	22	0	0	0	0	1	3	12	5	1	0	0	0	(11)
(12)		CDH23.3	97	0	0	0	0	3	21	57	14	2	1	0	0	(12)
(13)		CDH26.7	13	0	0	0	0	0	2	9	2	0	0	0	0	(13)
(14)	Wind	WSAvg	5.0	5.7	5.5	5.3	4.9	5.1	4.6	4.4	4.4	4.5	5.2	5.2	5.6	(14)
(15)	Precipitation	PrecAvg	947	92	64	71	56	61	62	70	82	90	97	105	98	(15)
(16)		PrecMax	1274	172	136	159	121	154	151	168	201	224	213	215	209	(16)
(17)		PrecMin	692	9	1	13	10	5	5	13	23	26	11	39	31	(17)
(18)		PrecStd	126	42	35	35	29	36	36	39	44	47	49	47	42	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.3	12.8	15.9	20.5	23.4	26.4	27.8	25.1	23.1	18.8	15.1	13.8	(19)
(20)			MCWB	10.9	10.8	11.1	13.7	16.7	18.3	18.8	19.0	17.5	15.9	13.6	12.1	(20)
(21)		2%	DB	11.3	11.2	13.3	17.3	20.2	22.7	24.9	22.1	20.8	17.2	14.2	12.6	(21)
(22)			MCWB	10.3	9.6	9.9	12.2	15.3	16.9	18.4	17.8	16.5	14.9	12.9	11.4	(22)
(23)		5%	DB	10.6	10.2	11.7	14.9	18.0	20.5	22.2	20.5	19.3	16.2	13.5	11.7	(23)
(24)			MCWB	9.7	8.9	9.2	11.2	14.0	15.8	17.1	16.8	15.9	14.3	12.3	10.6	(24)
(25)		10%	DB	9.9	9.5	10.6	13.2	16.1	18.8	20.5	19.4	18.2	15.5	12.7	10.9	(25)
(26)			MCWB	9.0	8.4	8.8	10.2	12.9	15.0	16.5	16.2	15.4	13.8	11.5	9.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	11.2	11.6	14.4	17.6	20.0	20.2	20.0	18.6	16.7	14.2	12.5	(27)
(28)			MCDWB	11.8	12.3	14.2	18.9	22.0	24.8	25.8	23.5	21.0	18.2	14.7	13.2	(28)
(29)		2%	WB	10.6	10.1	10.6	12.8	15.7	17.5	18.7	18.6	17.5	15.6	13.3	11.7	(29)
(30)			MCDWB	11.1	10.9	12.4	16.3	19.3	21.0	23.3	21.0	19.4	16.7	13.8	12.4	(30)
(31)		5%	WB	9.9	9.4	9.9	11.7	14.5	16.4	17.8	17.8	16.7	14.9	12.5	10.9	(31)
(32)			MCDWB	10.4	10.1	11.2	14.3	17.3	19.6	21.3	19.4	18.5	15.9	13.2	11.6	(32)
(33)		10%	WB	9.2	8.7	9.2	10.7	13.5	15.6	17.1	17.1	16.0	14.2	11.7	10.0	(33)
(34)			MCDWB	9.9	9.4	10.4	12.8	15.6	18.2	19.5	18.7	17.7	15.3	12.5	10.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.4	4.9	5.7	6.7	6.5	6.3	6.2	5.9	6.3	5.3	4.8	4.5	(35)
(36)			MCDBR	4.1	4.9	7.1	9.5	9.5	9.7	9.7	7.5	7.6	5.4	4.5	4.4	(36)
(37)			MCWBR	4.1	4.0	4.8	5.7	5.7	5.3	4.9	4.5	4.6	4.0	4.0	4.4	(37)
(38)		5% WB	MCDWB	4.2	4.5	6.0	7.9	8.2	8.1	8.0	6.2	6.2	4.7	4.2	4.3	(38)
(39)			MCWBR	4.4	4.0	4.5	5.2	5.3	5.0	4.6	4.3	4.5	4.1	4.1	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.330	0.344	0.375	0.392	0.398	0.388	0.387	0.374	0.366	0.354	0.328	0.320	(40)	
(41)		taud	2.354	2.383	2.304	2.277	2.301	2.360	2.366	2.422	2.435	2.438	2.408	2.329	(41)	
(42)		Ebn at Noon	674	776	816	845	857	867	861	856	820	755	674	617	(42)	
(43)		Edh at Noon	64	81	105	120	123	117	114	103	91	76	60	56	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.79	1.48	2.63	4.05	4.97	5.40	5.06	4.19	3.16	1.79	0.95	0.61	(44)	
(45)		RadStd	0.07	0.18	0.34	0.45	0.25	0.57	0.47	0.36	0.23	0.15	0.09	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page